1.11 Maclaurin Series

Worked Solutions

1 a)

$$e^{3x} = 1 + 3x + \frac{(3x)^2}{2!} + \frac{(3x)^3}{3!} + \cdots$$
$$\approx 1 + 3x + \frac{9}{2}x^2 + \frac{9}{2}x^3 \text{ valid for all } x$$

1 b)

$$e^{-2x} = 1 - 2x + \frac{(-2x)^2}{2!} + \frac{(-2x)^3}{3!} + \cdots$$
$$\approx 1 - 2x + 2x^2 - \frac{4}{3}x^3 \text{ valid for all } x$$

1 c)

Validity: $-1 < 2x \leq 1 \Rightarrow -\frac{1}{2} < x \leq \frac{1}{2}$

$$\ln(1 + 2x) = 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \frac{(2x)^4}{4} + \cdots$$
$$\approx 2x - 2x^2 + \frac{8}{3}x^3 - 4x^4 \text{ valid for } -\frac{1}{2} < x \leq \frac{1}{2}$$

1 d)

Validity: $-1 < -x \leq 1 \Rightarrow -1 \leq x < 1$

$$\ln(1 - x) = -x - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \frac{(-x)^4}{4} + \cdots$$
$$\approx -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 \text{ valid for } -1 \leq x < 1$$

1 e)

$$\sin 2x = 2x - \frac{(2x)^3}{3!} - \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!} + \cdots$$
$$\approx 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{315}x^7 \text{ valid for all } x$$

1 f)

$$\sin(-4x) = -4x - \frac{(-4x)^3}{3!} - \frac{(-4x)^5}{5!} - \frac{(-4x)^7}{7!} + \cdots$$
$$\approx -4x - \frac{32}{3}x^3 - \frac{128}{15}x^5 + \frac{1024}{315}x^7 \text{ valid for all } x$$

1 g)

$$\cos\left(\frac{1}{2}x\right) = 1 - \frac{(\frac{1}{2}x)^2}{2!} + \frac{(\frac{1}{2}x)^4}{4!} - \frac{(\frac{1}{2}x)^6}{6!} + \cdots$$
$$\approx 1 - \frac{1}{8}x^2 + \frac{1}{384}x^4 - \frac{1}{46080}x^6 \text{ valid for all } x$$

1 h)

$$\cos(-x) = 1 - \frac{(-x)^2}{2!} + \frac{(-x)^4}{4!} - \frac{(-x)^6}{6!} + \cdots$$
$$\approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 \text{ valid for all } x$$

2 a)

Second term: $-\frac{(\frac{5}{2}x)^2}{2!}$

Coefficient: $-\frac{25}{8}$

2 b)

Second term: $-4x$

Coefficient: -4

2 c)

Second term: $-\frac{(\frac{1}{2}x)^2}{2}$

Coefficient: $-\frac{1}{8}$

3 a)

$$\ln(3+x) = \ln\left(3\left(1+\frac{1}{3}x\right)\right) = \ln 3 + \ln\left(1+\frac{1}{3}x\right)$$
$$\ln(1-x) \approx \ln 3 + \frac{1}{3}x - \frac{(\frac{1}{3}x)^2}{2} + \cdots$$
$$\approx \ln 3 + \frac{1}{3}x - \frac{1}{18}x^2$$

3 b)

Valid for $-1 < \frac{1}{3}x \leq 1$

So $-3 < x \leq 3$

4 a)

$$\ln\left(1+\frac{1}{4}x\right) = \left(\frac{1}{4}x\right) - \frac{(\frac{1}{4}x)^2}{2} + \frac{(\frac{1}{4}x)^3}{3} + \cdots$$
$$\approx \frac{1}{4}x - \frac{1}{32}x + \frac{1}{192}x^3$$

4 b)

$-1 < \frac{1}{4}x \leq 1$ so $-4 < x \leq 4$

4 c)

$$\ln\sqrt{1+\frac{1}{4}x} = \ln\left(1+\frac{1}{4}x\right)^{\frac{1}{2}} = \frac{1}{2}\ln\left(1+\frac{1}{3}x\right)$$
$$\approx \frac{1}{8}x - \frac{1}{64}x + \frac{1}{384}x^3$$

5)

$$\frac{p^5}{5!} = \frac{4}{15} \Rightarrow p^5 = 32$$
$$\therefore p = 2$$

6)

$$\frac{p^2}{2!} = 72$$
$$p^2 = 144$$

So $p = \pm 12$

7 a)

$$\ln(4(1+x)) = \ln 4 + \ln(1+x)$$
$$\approx \ln 4 + x - \frac{x^2}{2} + \frac{x^3}{3}$$

7 b)

$$\frac{3e^x+1}{e^x} = 3 + e^{-x}$$
$$\approx 3 + 1 + (-x) + \frac{(-x)^2}{2!} + \frac{(-x)^3}{3!}$$
$$\approx 4 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3$$

7 c)

$$\ln(e - 2ex) = \ln(e(1-2x))$$
$$\approx \ln(e) + \ln(1-2x)$$
$$\approx \ln(e) + (-2x) - \frac{(-2x)^2}{2} + \frac{(-2x)^3}{3}$$
$$\approx 1 - 2x - 2x^2 - \frac{8}{3}x^3$$

7 d)

$$\ln(1+x^2) = x^2 - \frac{(x^2)^2}{2!} + \frac{(x^2)^3}{3!} \approx x^2 - \frac{x^4}{2} + \frac{x^2}{6}$$
$$\frac{1}{x}\ln(1+x^2) \approx x - \frac{x^3}{2}$$

7 e)

$$e^{x^2} = 1 + (x^2) + \frac{(x^2)^2}{2!} + \cdots \approx 1 + x^2 + \frac{1}{2}x^4$$
$$\frac{1}{x}e^{x^2} \approx \frac{1}{x} + x + \frac{1}{2}x^3$$

7 f)

$$\ln(2-2x) = \ln(2(1-x)) = \ln 2 + \ln(1-x)$$
$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$$
$$\ln(2-2x) = \ln 2 - x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$$
$$\frac{1}{x}\ln(2-2x) = \frac{\ln 2}{x} - 1 - \frac{x}{2} - \frac{x^2}{3} - \frac{x^3}{4} - \cdots$$
$$\approx \frac{\ln 2}{x} - 1 - \frac{x}{2} - \frac{x^2}{3} - \frac{x^3}{4}$$

8 a)

$$\cos(2x) \approx 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \frac{(2x)^8}{8!}$$
$$\approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 + \frac{2}{315}x^8$$

8 b)

$$\cos(-2x) \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6 + \frac{2}{315}x^8$$

9 a)

$$\sin(2x) \approx 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!}$$
$$\approx 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{315}x^7$$

9 b)

$$\sin(-2x) \approx -2x + \frac{4}{3}x^3 - \frac{4}{15}x^5 + \frac{8}{315}x^7$$

10 a)

$$(4+4x)^{\frac{1}{2}} = (4(1+x))^{\frac{1}{2}} = 4^{\frac{1}{2}}(1+x)^{\frac{1}{2}}$$
$$= 2\left[1 + \frac{1}{2}x + \frac{(\frac{1}{2})(-\frac{1}{2})x^2}{2} + \cdots\right]$$
$$= 2 + x - \frac{1}{4}x^2 + \cdots \text{ for } |x| < 1$$

10 b)

$$(4+x)^{\frac{1}{2}} = \left(4\left(1+\frac{1}{4}x\right)\right)^{\frac{1}{2}} = 4^{\frac{1}{2}}\left(1+\frac{1}{4}x\right)^{\frac{1}{2}}$$
$$= 2\left[1 + \frac{1}{2}\left(\frac{1}{4}x\right) + \frac{(\frac{1}{2})(-\frac{1}{2})(\frac{1}{4}x)^2}{2} + \cdots\right]$$
$$= 2 + \frac{1}{4}x - \frac{1}{64}x^2 + \cdots \text{ for } \left|\frac{1}{4}x\right| < 1$$

So $(4+x)^{\frac{1}{2}} \approx 2 + \frac{1}{4}x - \frac{1}{64}x^2$ for $|x| < 4$

10 c)

$$(4+5x)^{-\frac{1}{2}} = \left(4\left(1+\frac{5}{4}x\right)\right)^{-\frac{1}{2}} = 4^{-\frac{1}{2}}\left(1+\frac{5}{4}x\right)^{-\frac{1}{2}}$$
$$= \frac{1}{2}\left[1 + \left(-\frac{1}{2}\right)\left(\frac{5}{4}x\right) + \frac{(-\frac{1}{2})(-\frac{3}{2})(\frac{5}{4}x)^2}{2} + \cdots\right]$$
$$= \frac{1}{2} - \frac{5}{16}x + \frac{75}{256}x^2 + \cdots \text{ for } \left|\frac{5}{4}x\right| < 1$$

So $(4+5x)^{-\frac{1}{2}} \approx \frac{1}{2} - \frac{5}{16}x + \frac{75}{256}x^2$ for $|x| < \frac{4}{5}$

11 a)

$$\ln(1+3x^2-x) = (3x^2-x) - \frac{(3x^2-x)^2}{2} + \frac{(3x^2-x)^3}{3}\cdots$$
$$= 3x^2 - x - \frac{9x^4-6x^3+x^2}{2} + \frac{(3x^2-x)(9x^4-6x^3+x^2)}{3}\cdots$$
$$= 3x^2 - x + 3x^3 - \frac{1}{2}x^2 - \frac{1}{3}x^3\cdots$$
$$\approx -x + \frac{5}{2}x^2 + \frac{8}{3}x^3$$
$$-1 < 3x^2 - x \leq 1$$
$$3x^2 - x \leq 1 \Rightarrow 3x^2 - x - 1 \leq 0 \Rightarrow \frac{1}{6} - \frac{\sqrt{13}}{6} \leq x \leq \frac{1}{6} + \frac{\sqrt{13}}{6}$$
$$-1 < 3x^2 - x \Rightarrow 3x^2 - x + 1 > 0 \Rightarrow 3\left(x - \frac{1}{6}\right)^2 + \frac{11}{12} > 0$$

which is always true

So $\ln(1+3x^2-x) \approx -x + \frac{5}{2}x^2 + \frac{8}{3}x^3$ where $\frac{1}{6} - \frac{\sqrt{13}}{6} \leq x \leq \frac{1}{6} + \frac{\sqrt{13}}{6}$

11 b)

$$\ln(1+2x^2+3x) = (2x^2+3x) - \frac{(2x^2+3x)^2}{2} + \frac{(2x^2+3x)^3}{3} + \cdots$$
$$= 2x^2+3x - \frac{4x^4+12x^3+9x^2}{2} + \frac{(2x^2+3x)(4x^4+12x^3+9x^2)}{3} + \cdots$$
$$= 2x^2+3x-6x^3-\frac{9}{2}x^2+9x^3\ldots$$
$$\approx 3x-\frac{5}{2}x^2+3x^3$$
$$-1 < 2x^2+3x \leq 1$$
$$2x^2+3x \leq 1 \Rightarrow 2x^2+3x-1 \leq 0 \Rightarrow -\frac{3}{4}-\frac{\sqrt{17}}{4} \leq x \leq -\frac{3}{4}+\frac{\sqrt{17}}{4}$$
$$-1 < 2x^2+3x \Rightarrow 2x^2+3x+1 > 0 \Rightarrow x > -\frac{1}{2},\ x < -1$$

So $\ln(1+2x^2+3x) \approx 3x-\frac{5}{2}x^2+3x^3$

where $-\frac{3}{4}-\frac{\sqrt{17}}{4} \leq x < -1$ or $-\frac{1}{2} < x \leq -\frac{3}{4}+\frac{\sqrt{17}}{4}$

12 a)

$$\ln\left(1+\frac{1}{2}x^2+4x\right) = \left(\frac{1}{2}x^2+4x\right) - \frac{(\frac{1}{2}x^2+4x)^2}{2} + \frac{(\frac{1}{2}x^2+4x)^3}{3} + \cdots$$
$$\frac{1}{2}x^2+4x - \frac{\frac{1}{4}x^4+4x^3+16x^2}{2} + \frac{(\frac{1}{2}x^2+4x)(\frac{1}{4}x^4+4x^3+16x^2)}{3} + \cdots$$
$$\frac{1}{2}x^2+4x-2x^3-8x^2+\frac{64}{3}x^3+\cdots$$
$$\approx 4x-\frac{15}{2}x^2+\frac{58}{3}x^3$$

12 b)

$$\ln(x^2+8x+2) = \ln\left(2\left(\frac{1}{2}x^2+4x+1\right)\right)$$
$$= \ln 2 + \ln\left(1+\frac{1}{2}x^2+4x\right)$$
$$\approx \ln 2 + 4x - \frac{15}{2}x^2+\frac{58}{3}x^3$$

13 a)

$$e^x+e^{-x} = 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$$
$$+1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$$
$$= 1+x+\frac{x^2}{2}+\frac{x^3}{6}+\frac{x^4}{24}+\cdots$$
$$+1-x+\frac{x^2}{2}-\frac{x^3}{6}+\frac{x^4}{24}+\cdots$$
$$\approx 2+x^2+\frac{1}{12}x^4$$

13 b)

$$\begin{aligned}\sin x + \cos x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \\ &\approx 1 + x - \frac{x^2}{2}\end{aligned}$$

13 c)

$$\begin{aligned}\cos x - \sin x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots - x + \frac{x^3}{3!} - \frac{x^5}{5!} - \cdots \\ &\approx 1 - x - \frac{x^2}{2}\end{aligned}$$

13 d)

$$\begin{aligned}e^x - \sin x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots - x + \frac{x^3}{3!} - \frac{x^5}{5!} + \cdots \\ &\approx 1 + \frac{x^2}{2} + \frac{x^3}{3}\end{aligned}$$

13 e)

$$\begin{aligned}e^x - \cos x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots - 1 + \frac{x^2}{2!} - \frac{x^4}{4!} - \cdots \\ &\approx x + x^2 + \frac{x^3}{6}\end{aligned}$$

13 f)

$$\begin{aligned}2\sin 2x + 2\cos 2x &= 2\left(2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots\right) + 2\left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots\right) \\ &= 2\left(2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 - \cdots\right) + 2\left(1 - 2x^2 + \frac{2}{3}x^4 - \cdots\right) \\ &= 4x - \frac{8}{3}x^3 + \frac{8}{15}x^5 - \cdots + 2 - 4x^2 + \frac{4}{3}x^4 - \cdots \\ &\approx 2 + 4x - 4x^2\end{aligned}$$

13 g)

$$\begin{aligned}e^{ax} + e^{-ax} &= 1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \frac{(ax)^4}{4!} + \cdots \\ &\quad + 1 - ax + \frac{(ax)^2}{2!} - \frac{(ax)^3}{3!} + \frac{(ax)^4}{4!} + \cdots \\ &= 1 + ax + \frac{a^2x^2}{2} + \frac{a^3x^3}{6} + \frac{a^4x^4}{24} + \cdots \\ &\quad + 1 - ax + \frac{a^2x^2}{2} - \frac{a^3x^3}{6} + \frac{a^4x^4}{24} + \cdots \\ &\approx 2 + a^2x^2 + \frac{a^4}{12}x^4\end{aligned}$$

13 h)

$$\begin{aligned}\sin ax + x\cos ax &= ax - \frac{(ax)^3}{3!} + \frac{(ax)^5}{5!} - \cdots + x\left(1 - \frac{(ax)^2}{2!} + \frac{(ax)^4}{4!} - \cdots\right)\\ &= ax - \frac{a^3}{6}x^3 + \frac{(ax)^5}{5!}\ldots + x\left(1 - \frac{a^2}{2}x^2 + \frac{a^4}{24}x^4 - \cdots\right)\\ &= ax - \frac{a^3}{6}x^3 + \frac{a^5}{120}x^5\ldots + x - \frac{a^2}{2}x^3 + \frac{a^4}{24}x^5 - \cdots\\ &\approx (1+a)x - \frac{a^2}{6}(a+3)x^3 + \frac{1}{120}a^4(a+5)x^5\end{aligned}$$

14 a)

$$|x| < \frac{1}{b}$$

14 b)

$$\begin{aligned}\mathrm{f}(x) &= 1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \cdots\\ &\quad + 1 + \left(\frac{1}{2}\right)(bx) + \frac{(\frac{1}{2})(-\frac{1}{2})(bx)^2}{2} + \frac{(\frac{1}{2})(-\frac{1}{2})(-\frac{3}{2})(bx)^3}{6} + \cdots\\ &\approx 1 + ax + \frac{a^2}{2}x^2 + \frac{a^3}{6}x^3 + 1 + \frac{b}{2}x - \frac{b^2}{8}x^2 + \frac{b^3}{16}x^3\\ &= 2 + \left(a + \frac{b}{2}\right)x + \left(\frac{a^2}{2} - \frac{b^2}{8}\right)x^2 + \left(\frac{a^3}{6} + \frac{b^3}{16}\right)x^3\end{aligned}$$

14 c)

$a + \frac{b}{2} = -1$ and $\frac{a^2}{2} - \frac{b^2}{8} = \frac{5}{2}$

So $b = -2a - 2$ which gives

$$\begin{aligned}\frac{a^2}{2} - \frac{(-2a-2)^2}{8} = \frac{5}{2} &\Rightarrow 4a^2 - (-2a-2)^2 = 20\\ 4a^2 - (4a^2 + 8a + 4) &= 20\\ -8a &= 24\end{aligned}$$

so $a = -3$ and $b = 4$

14 d)

$$\frac{a^3}{6} + \frac{b^3}{48} = \frac{(-3)^3}{6} + \frac{4^3}{16} = -\frac{1}{2}$$

15)

$$
\begin{aligned}
e^{2x} &= 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \frac{(2x)^5}{5!} + \cdots \\
&= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots \\
e^{-2x} &= 1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 - \frac{4}{15}x^5 + \cdots \\
\sinh(2x) &= \frac{1}{2}(e^{2x} - e^{-2x}) \\
&= \frac{1}{2}\Bigg(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots \\
&\quad - \left(1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 - \frac{4}{15}x^5 + \cdots\right)\Bigg) \\
&= \frac{1}{2}\Bigg(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots \\
&\quad -1 + 2x - 2x^2 + \frac{4}{3}x^3 - \frac{2}{3}x^4 + \frac{4}{15}x^5 + \cdots\Bigg) \\
&\approx \frac{1}{2}\left(4x + \frac{8}{3}x^3 + \frac{8}{15}x^5\right) \\
&\approx 2x + \frac{4}{3}x^3 + \frac{4}{15}x^5
\end{aligned}
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16)

$$
\begin{aligned}
e^{2x} &= 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \cdots \\
&= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \cdots \\
e^{-2x} &= 1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 + \cdots \\
\cosh(2x) &= \frac{1}{2}(e^{2x} - e^{-2x}) \\
&= \frac{1}{2}\Bigg(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \cdots \\
&\quad +1 - 2x + 2x^2 - \frac{4}{3}x^3 + \frac{2}{3}x^4 - \cdots\Bigg) \\
&\approx \frac{1}{2}\left(2 + 4x^2 + \frac{4}{3}x^4\right) \\
&\approx 1 + 2x^2 + \frac{2}{3}x^4
\end{aligned}
$$

17 a)

$$
\begin{aligned}
e^{ix} &= 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \cdots \\
&= 1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots \\
e^{-ix} &= 1 - ix + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \cdots \\
&= 1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots \\
\cosh(ix) &= \frac{1}{2}(e^{ix} + e^{-ix}) \\
&= \frac{1}{2}\Bigg(1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots \\
&\quad +1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 + \cdots\Bigg) \\
&= \frac{1}{2}\left(2 - x^2 + \frac{1}{12}x^4 + \cdots\right) \\
&\approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4
\end{aligned}
$$

17 b)

$$
1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 = 1 - \frac{x^2}{2!} + \frac{x^4}{6!} = \cos x
$$

$$
\therefore \cos x
$$

18 a)

$$e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \cdots$$
$$= 1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots$$
$$e^{-ix} = 1 - ix + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \frac{(-ix)^5}{5!} + \cdots$$
$$= 1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 - \frac{i}{120}x^5 + \cdots$$
$$\sinh(ix) = \frac{1}{2}(e^{ix} + e^{-ix})$$
$$= \frac{1}{2}\Bigg(1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots$$
$$- \left(1 - ix - \frac{1}{2}x^2 + \frac{i}{6}x^3 + \frac{1}{24}x^4 - \frac{i}{120}x^5 + \cdots\right)\Bigg)$$
$$= \frac{1}{2}\Bigg(1 + ix - \frac{1}{2}x^2 - \frac{i}{6}x^3 + \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots$$
$$-1 + ix + \frac{1}{2}x^2 - \frac{i}{6}x^3 - \frac{1}{24}x^4 + \frac{i}{120}x^5 + \cdots\Bigg)$$
$$\approx \frac{1}{2}\left(2ix - \frac{i}{3}x^3 + \frac{i}{60}x^5\right)$$
$$\approx ix - \frac{i}{6}x^3 + \frac{i}{120}x^5$$

18 b)

$$ix - \frac{i}{6}x^3 + \frac{i}{120}x^5 = i\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5\right)$$
$$\therefore \sin x$$

19 a)

$$(1+2x)^{\frac{1}{3}} = 1 + \left(\frac{1}{3}\right)(2x) + \frac{(\frac{1}{3})(-\frac{2}{3})(2x)^2}{2} + \frac{(\frac{1}{3})(-\frac{2}{3})(-\frac{5}{3})(2x)^3}{6} + \cdots$$
$$= 1 + \frac{2}{3}x - \frac{4}{9}x^2 + \frac{40}{81}x^3 + \cdots$$
$$\ln(1+5x) = (5x) - \frac{(5x)^2}{2} + \frac{(5x)^3}{3} + \cdots = 5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 + \cdots$$
$$k(1+2x)^{\frac{1}{3}} - \left(1 + \frac{2}{3}x - \frac{4}{9}x^2 + \frac{40}{81}x^3 + \cdots\right)$$
$$= k\left(1 + \frac{2}{3}x - \frac{4}{9}x^2 + \frac{40}{81}x^3 + \cdots\right) - \left(5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 + \cdots\right)$$

$\frac{2}{3}k - 5 = -1$ so $k = 6$

19 b)

$$6\left(1+\frac{2}{3}x-\frac{4}{9}x^2+\frac{40}{81}x^3+\cdots\right)-\left(5x-\frac{25}{2}x^2+\frac{125}{3}x^3+\cdots\right)$$
$$=6+4x-\frac{8}{3}x^2+\frac{80}{27}x^3+\cdots-5x+\frac{25}{2}x^2-\frac{125}{3}x^3-\cdots$$
$$\approx 6-x+\frac{59}{6}x^2-\frac{1045}{27}x^3$$

19 c)

$(1+2x)^{\frac{1}{3}}$ is valid for $-\frac{1}{2}<x\leq\frac{1}{2}$

$\ln(1+5x)$ is valid for $-\frac{1}{5}<x\leq\frac{1}{5}$

So f(x) is valid for $-\frac{1}{5}<x\leq\frac{1}{5}$

20 a)

$$(1-2x-8x^2)=(1+2x)(1-4x)$$

20 b)

$$\ln(1-2x-8x^2)$$
$$=\ln((1+2x)(1-4x))$$
$$=\ln(1+2x)+\ln(1-4x)$$
$$=\left(2x-2x^2+\frac{8}{3}x^3-\right)\ldots+\left(-4x-8x^2-\frac{64}{3}x^3-\cdots\right)$$
$$\approx -2x-10x^2-\frac{56}{3}x^3$$

20 c)

$\ln(1+2x)$ is valid for $-1<2x\leq 1\therefore -\frac{1}{2}<x\leq\frac{1}{2}$

$\ln(1-4x)$ is valid for $-1<-4x\leq 1$

$-1<-4x$ gives $x<\frac{1}{4}$ and $-4x\leq 1$ gives $x\geq-\frac{1}{4}$ so $-\frac{1}{4}\leq x<\frac{1}{4}$

So $\ln(1-2x-8x^2)$ is valid for $-\frac{1}{4}\leq x<\frac{1}{4}$

21 a)

$$e^x\sin x=\left(1+x+\frac{x^2}{2!}+\cdots\right)\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots\right)$$
$$=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots+x^2-\frac{x^4}{3!}+\frac{x^6}{5!}-\cdots+\frac{x^3}{2}-\frac{x^5}{2\times 3!}+\frac{x^7}{2\times 5!}+\cdots$$
$$\approx x+x^2-\frac{x^3}{3!}+\frac{x^3}{2}+\cdots$$
$$\approx x+x^2+\frac{1}{3}x^3$$

21 b)

$$e^x \cos x = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right)$$
$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + x - \frac{x^3}{2!} + \frac{x^5}{4!} - \cdots + \frac{x^2}{2!} - \frac{x^4}{2 \times 2!} + \frac{x^6}{2 \times 4!} - \cdots$$
$$\approx 1 + x - \frac{1}{2}x^3$$

21 c)

$$\sin x \ln(1+x) = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots\right)$$
$$= x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \cdots - \frac{x^4}{6} + \frac{x^5}{6 \times 2} - \frac{x^6}{6 \times 3} + \cdots$$
$$\approx x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^4}{6} \cdots$$
$$\approx x^2 - \frac{1}{2}x^3 + \frac{1}{6}x^4$$

21 d)

$$e^{-x} \sin x = \left(1 - x + \frac{x^2}{2!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$$
$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots - x^2 + \frac{x^4}{3!} - \frac{x^6}{5!} - \cdots + \frac{x^3}{2} - \frac{x^5}{2 \times 3!} + \frac{x^7}{2 \times 5!} + \cdots$$
$$\approx x - \frac{1}{6}x^3 - x^2 + \frac{1}{2}x^3$$
$$\approx x - x^2 + \frac{1}{3}x^3$$

21 e)

$$e^x \cos 2x = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots\right)$$
$$= \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(1 - 2x^2 + \frac{2}{3}x^4 - \cdots\right)$$
$$= 1 - 2x^2 + \frac{2}{3}x^4 - \cdots + x - 2x^3 + \frac{2}{3}x^5 - \cdots + \frac{1}{2}x^2 - x^4 + \frac{2}{3}x^6 - \cdots$$
$$\approx 1 + x - \frac{3}{2}x^2$$

22 a)

$$(3+5x)^{-1} = \left(3\left(1+\frac{5}{3}x\right)\right)^{-1}$$
$$\approx \frac{1}{3}\left[1+(-1)\left(\frac{5}{3}x\right)+\frac{(-1)(-2)(\frac{5}{3}x)^2}{2}+\frac{(-1)(-2)(-3)(\frac{5}{3}x)^3}{6}+\cdots\right]$$
$$= \frac{1}{3}\left[1-\frac{5}{3}x+\frac{25}{9}x^2-\frac{125}{27}x^3+\cdots\right]$$
$$\approx \frac{1}{3}-\frac{5}{9}x+\frac{25}{27}x^2-\frac{125}{81}x^3$$

22 b)

Valid for $\left|\frac{5}{3}x\right| < 1$ so $|x| < \frac{3}{5}$

22 c)

$$\sin 2x = 2x-\frac{(2x)^3}{3!}+\frac{(2x)^5}{5!}-\cdots$$
$$= 2x-\frac{4}{3}x^3+\frac{4}{15}x^5-\cdots$$

So that

$$(\sin 2x)(3+5x)^{-1} = \left(2x-\frac{4}{3}x^3+\frac{4}{15}x^5-\cdots\right)\left(\frac{1}{3}-\frac{5}{9}x+\frac{25}{27}x^2-\frac{125}{81}x^3\right)$$
$$= \frac{2}{3}x-\frac{10}{9}x^2+\frac{50}{27}x^3-\frac{250}{81}x^4+\cdots-\frac{4}{9}x^3+\frac{20}{27}x^4+\cdots$$
$$\approx \frac{2}{3}x-\frac{10}{9}x^2+\frac{38}{27}x^3-\frac{190}{81}x^4$$

22 d)

$$|x| < \frac{3}{5}$$

23 a)

$$\cos\theta\sin\theta = \left(1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-\cdots\right)\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-\cdots\right)$$
$$= \theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-\cdots-\frac{\theta^3}{2}+\frac{\theta^5}{2\times 3!}-\frac{\theta^7}{2\times 5!}-\cdots$$
$$+\frac{\theta^5}{24}-\frac{\theta^7}{24\times 3!}+\frac{\theta^9}{24\times 5!}-\cdots$$
$$\approx \theta-\frac{\theta^3}{6}+\frac{\theta^5}{120}-\frac{\theta^3}{2}+\frac{\theta^5}{12}+\frac{\theta^5}{24}$$
$$\approx \theta-\frac{2}{3}\theta^3+\frac{2}{15}\theta^5$$
$$\sin q\theta = q\theta-\frac{(q\theta)^3}{3!}-\frac{(q\theta)^5}{5!}+\cdots$$
$$\approx q\theta-\frac{q^3}{6}\theta^3+\frac{q^5}{120}\theta^5$$

So $\cos\theta\sin\theta \approx \theta - \frac{2}{3}\theta^3 + \frac{2}{15}\theta^5$ and $\sin q\theta \approx q\theta - \frac{q^3}{6}\theta^3 + \frac{q^5}{120}\theta^5$

23 b)

$$\theta - \frac{2}{3}\theta^3 + \frac{2}{15}\theta^5 = pq\theta - \frac{pq^3}{6}\theta^3 + \frac{pq^5}{120}\theta^5$$
$$pq = 1$$
$$-\frac{2}{3} = -\frac{pq^3}{6} \Rightarrow 4 = pq^3 \Rightarrow 4 = q^2 \therefore q = 2$$

$p = \frac{1}{2}$ and $q = 2$

24)

$$e^{px}\sin x = \left(1 + px + \frac{p^2x^2}{2!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$$
$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + px^2 - \frac{px^4}{3!} + \frac{px^6}{5!} - \cdots + \frac{p^2}{2}x^3 - \cdots$$

So the third term is

$$-\frac{1}{6}x^3 + \frac{p^2}{2}x^3$$
$$\frac{p^2}{2} - \frac{1}{6} = \frac{13}{3} \Rightarrow p^2 = 9$$

$\therefore p = -3$ or $p = 3$

25)

$$e^{x}\sin px = \left(1 + x + \frac{x^2}{2!} + \cdots\right)\left(px - \frac{p^3x^3}{3!} + \frac{p^5x^5}{5!} - \cdots\right)$$
$$= px - \frac{p^3x^3}{3!} + \frac{p^5x^5}{5!} - \cdots + px^2 - \frac{p^3x^4}{3!} + \frac{p^5x^6}{5!} - \cdots + \frac{p}{2!}x^3 - \cdots$$

So the third term is

$$-\frac{p^3}{6}x^3 + \frac{p}{2}x^3$$
$$-\frac{p^3}{6} + \frac{p}{2} = -\frac{1}{3} \Rightarrow p^3 - 3p = 2 \Rightarrow p^3 - 3p - 2 = 0 \Rightarrow (p-2)(p+1)^2 = 0$$

$\therefore p = 2$ or $p = -1$

26)

$$\frac{\cos px}{e^x} = e^{-x}\cos px = \left(1 - x + \frac{x^2}{2!} + \cdots\right)\left(1 - \frac{(px)^2}{2!} + \frac{(px)^4}{4!} - \cdots\right)$$
$$= 1 - \frac{p^2x^2}{2!} + \frac{p^4x^4}{4!} - \cdots - x + \frac{p^2x^3}{2!} + \cdots + \frac{x^2}{2!} - \cdots$$

So the third term is

$$-\frac{p^2x^2}{2} + \frac{x^2}{2}$$
$$-\frac{p^2}{2} + \frac{1}{2} = -12 \Rightarrow p^2 - 1 = 24$$

$\therefore p = -5$ or $p = 5$

27)

$$\frac{\sin ax}{be^x} = \frac{1}{b}\sin ax\, e^{-x}$$
$$= \frac{1}{b}\left(ax - \frac{a^3x^3}{3!} + \frac{a^5x^5}{5!} - \cdots\right)\left(1 - x + \frac{x^2}{2!} + \cdots\right)$$
$$= \frac{1}{b}\left(ax - ax^2 + \frac{ax^3}{2} + \cdots - \frac{a^3x^3}{6} + \frac{a^3x^4}{6} - \frac{a^3x^5}{12} + \cdots\right)$$

$\therefore \frac{a}{b} = \frac{1}{2}$ and $\frac{1}{b}\left(\frac{a}{2} - \frac{a^3}{6}\right) = -\frac{1}{12}$

$$\frac{a}{2b} - \frac{a^3}{6b} = -\frac{1}{12} \Rightarrow \frac{1}{4} - \frac{a^2}{12} = -\frac{1}{12} \Rightarrow a^2 = 4$$

The second term is $-\frac{a}{b}$

$\therefore a = 2$ and $b = 4$ and $k = \frac{1}{2}$

28 a)

$$e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2!} + \frac{(0.1)^3}{3!} + \cdots$$
$$\approx 1 + 0.1 + 0.005 + 0.0001666666 + \cdots$$
$$= 1.1052 \text{ to 4 decimal places}$$

28 b)

$$\ln 0.9 = \ln(1 - 0.1)$$
$$\approx -0.1 - \frac{(-0.1)^2}{2} + \frac{(-0.1)^3}{3} - \frac{(-0.1)^4}{4} + \cdots$$
$$\approx -0.1 - 0.005 - 0.00033333\ldots - 0.000025 + \cdots$$
$$= -0.1054 \text{ to 4 decimal places}$$

28 c)

$$\sin\left(\frac{\pi}{18}\right) \approx \frac{\pi}{18} - \frac{(\frac{\pi}{18})^3}{3!} + \cdots$$
$$\approx 0.17453292\ldots - 0.00088609\ldots + \cdots$$
$$= 0.1736 \text{ to 4 decimal places}$$

28 d)

$14^\circ = \frac{14\pi}{180}$ radians $= \frac{7\pi}{90}$ radians

$$\cos\left(\frac{7\pi}{90}\right) \approx 1 - \frac{(\frac{7\pi}{90})^2}{2!} + \frac{(\frac{7\pi}{90})^4}{4!} + \cdots$$
$$\approx 1 - 0.02985250\ldots + 0.00014852 + \cdots$$
$$= 0.9703 \text{ to 4 decimal places}$$

29 a)

$$0.993252\ldots$$

29 b)

The expansion is only valid for $-1 < x \leq 1$

29 c)

$$\begin{aligned}\ln(2.7) = \ln(3 \times 0.9) = \ln 3 + \ln 0.9 = \ln 3 + \ln(1 - 0.1)\\ \approx \ln 3 - 0.1 - \frac{(-0.1)^2}{2} + \frac{(-0.1)^3}{3} + \cdots\\ \approx 1.09861 - 0.1 - 0.005 - 0.00033333 + \cdots\\ = 0.9933 \text{ to 4 decimal places}\end{aligned}$$

30 a)

$$\begin{aligned}\frac{1}{\sqrt{e^x}} = e^{-\frac{1}{2}x} = 1 + \left(-\frac{1}{2}x\right) + \frac{(-\frac{1}{2}x)^2}{2!} + \frac{(-\frac{1}{2}x)^3}{3!} + \frac{(-\frac{1}{2}x)^4}{4!} + \frac{(-\frac{1}{5}x)^5}{5!}\cdots\\ \approx 1 - \frac{1}{2}x + \frac{1}{8}x^2 - \frac{1}{48}x^3 + \frac{1}{384}x^4 - \frac{1}{3840}x^5\cdots\end{aligned}$$

30 b)

$$\begin{aligned}\frac{1}{\sqrt{e^1}} \approx 1 - \frac{1}{2} + \frac{1}{8} - \frac{1}{48} + \frac{1}{384} - \frac{1}{3840} + \cdots\\ = 0.6065\end{aligned}$$

31 a)

$$\begin{aligned}e^x \approx 1 + x + \frac{x^2}{2!} \approx 1 + x + \frac{1}{2}x^2\\ \int_0^{0.5} e^x\,dx \approx \int_0^{0.5} \left(1 + x + \frac{1}{2}x^2\right)\,dx\\ = \left[x + \frac{1}{2}x^2 + \frac{1}{6}x^3\right]_0^{0.5}\\ = \left(0.5 + \frac{1}{2} \times 0.5^2 + \frac{1}{6} \times 0.5^3\right) - (0)\\ = 0.65 \text{ (2 DP)}\end{aligned}$$

31 b)

$$\ln(1+x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}$$

$$\int_0^{0.5} \ln(1+x)\,dx \approx \int_0^{0.5} \left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) dx$$

$$= \left[\frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4\right]_0^{0.5}$$

$$= \left(\frac{1}{2} \times 0.5^2 - \frac{1}{6} \times 0.5^3 + \frac{1}{12} \times 0.5^4\right) - (0)$$

$$= 0.11 \text{ (2 DP)}$$

31 c)

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} = x - \frac{1}{6}x^3 + \frac{1}{120}x^5$$

$$\int_0^{1.5} \sin x\,dx \approx \int_0^{1.5} \left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5\right) dx$$

$$= \left[\frac{1}{2}x^2 - \frac{1}{24}x^4 + \frac{1}{720}x^6\right]_0^{1.5}$$

$$= \left(\frac{1}{2} \times 1.5^2 - \frac{1}{24} \times 1.5^4 + \frac{1}{720} \times 1.5^6\right) - (0)$$

$$= 0.93 \text{ (2 DP)}$$

31 d)

$$\cos x \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$$

$$\int_0^{1.5} \cos x\,dx \approx \int_0^{1.5} \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4\right) dx$$

$$= \left[x - \frac{1}{6}x^3 + \frac{1}{120}x^5\right]_0^{1.5}$$

$$= \left(1.5 - \frac{1}{6} \times 1.5^3 + \frac{1}{120} \times 1.5^5\right) - (0)$$

$$= 1.00 \text{ (2 DP)}$$

31 e)

$$(1-x)^{\frac{1}{2}} \approx 1 - \frac{x}{2} - \frac{x^2}{8}$$

$$\int_0^{0.5} (1-x)^{\frac{1}{2}}\,dx \approx \int_0^{0.5} \left(1 - \frac{x}{2} - \frac{x^2}{8}\right) dx$$

$$= \left[x - \frac{1}{4}x^2 - \frac{1}{24}x^3\right]_0^{0.5}$$

$$= \left(0.5 - \frac{1}{4} \times 0.5^2 - \frac{1}{24} \times 0.5^4\right) - (0)$$

$$= 0.43 \text{ (2 DP)}$$

32)

$$(1-x)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)(-x) + \frac{(-\frac{1}{2})(-\frac{3}{2})(-x)^2}{2} + \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})(-x)^3}{6}$$
$$= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3$$
$$\therefore \int \frac{1}{\sqrt{1-x}}\,dx \approx \int \left(1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3\right) dx$$
$$= x + \frac{1}{4}x^2 + \frac{1}{8}x^3 + \frac{5}{64}x^4 + c$$

33 a)

$$\ln(1+2x) \approx (2x) - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} = 2x - 2x^2 + \frac{8}{3}x^3$$

33 b)

$2x - 2x^2 + \frac{8}{3}x^3 = 8x^2$ gives $\frac{8}{3}x^3 - 10x^2 + 2x = 0$

$x(4x^2 - 15x + 3) = 0$ so $x = 0$ or $x = \frac{15}{8} \pm \frac{\sqrt{177}}{8}$

Since only valid for $-\frac{1}{2} < x \leq \frac{1}{2}$

So $x = 0.212$ (3 SF)

33 c)

Include the quartic term in the expansion, as every term included improves the accuracy.

34 a)

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots$$
$$\approx 1 - \frac{1}{2}\theta^2 + \frac{1}{24}\theta^4$$

34 b)

$-\frac{1}{648}\theta^4 = 1 - \frac{1}{2}\theta^2 + \frac{1}{24}\theta^4$ gives $7\theta^4 - 81\theta^2 + 162 = 0$

$$(\theta^2 - 9)(7\theta^2 - 18) = 0$$

So solutions are $\theta = 3$, $\theta = -3$, $\theta = \pm\sqrt{\frac{18}{7}}$

So $\theta = 1.60$ (3 SF)

35 a)

$$\mathrm{f}(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots - \left(1 - \frac{x^2}{2!} - \cdots\right)$$
$$= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots - 1 + \frac{1}{2}x^2 + \cdots$$
$$\approx x + x^2 + \frac{1}{6}x^3$$

35 b)

$$x + x^2 + \frac{1}{6}x^3 = 0$$
$$\frac{1}{6}x(x^2 + 6x + 6) = 0$$

Has solutions $x = 0,\ -3 \pm \sqrt{3}$

Therefore $x = -1.27$ (3 SF)

36 a)

$$\ln\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$$
$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots$$
$$\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \cdots$$
$$\ln\left(\frac{x+1}{1-x}\right) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots - \left(-x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \cdots\right)$$
$$= 2x + \frac{2}{3}x^3 + \cdots$$

When $x = 0.2$

$$\ln\left(\frac{1+0.2}{1-0.2}\right) = \ln\left(\frac{1.2}{0.8}\right) = \ln(1.5) = 2(0.2) + \frac{2}{3}(0.2)^3 + \cdots$$
$$= 0.405333\ldots$$

So 0.405 (3 d.p.)

36 b)

$$\frac{\ln 1.5 - 0.405}{\ln 1.5} \times 100 = 0.115\%$$

37 a)

$$e^x\sqrt{1+2x} = e^x(1+2x)^{\frac{1}{2}}$$
$$= \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)\left(1 + \left(\frac{1}{2}\right)(2x)\right.$$
$$\left. + \frac{(\frac{1}{2})(-\frac{1}{2})(2x)^2}{2} + \frac{(\frac{1}{2})(-\frac{1}{2})(-\frac{3}{2})(2x)^3}{6} + \cdots\right)$$
$$= \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots\right)\left(1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \cdots\right)$$
$$= 1 + x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \cdots + x + x^2 - \frac{1}{2}x^3 + \cdots + \frac{1}{2}x^2 + \frac{1}{2}x^3 - \cdots + \frac{1}{6}x^3\ldots$$
$$\approx 1 + 2x + x^2 + \frac{2}{3}x^3$$

37 b)

$$1 + 2(0.5) + (0.5)^2 + \frac{2}{3}(0.5)^3 = 2.33 \text{ (3 SF)}$$

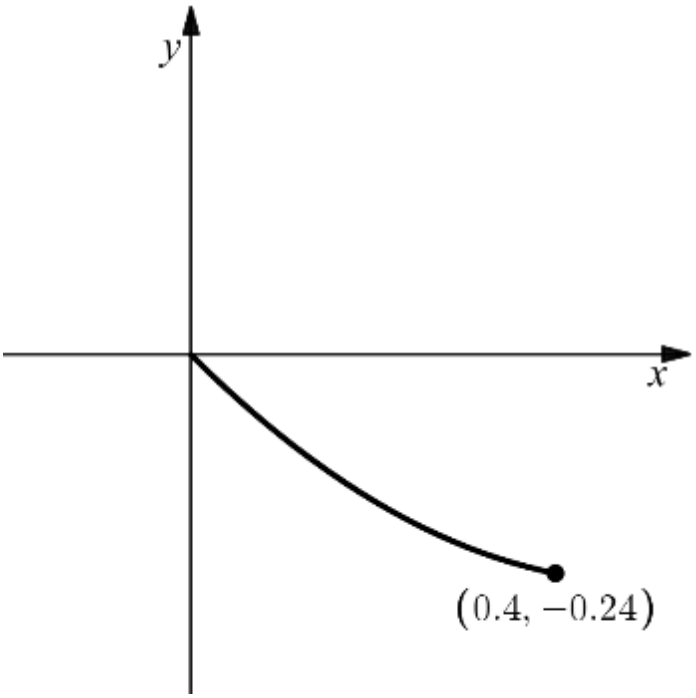
37 c)

$$\frac{\sqrt{2}e^{\frac{1}{2}} - 2.33}{\sqrt{2}e^{\frac{1}{2}}} \times 100 = 0.0705\%$$

38)

$$e^{-x} \approx 1 - x + \frac{x^2}{2!}$$
$$\cos x \approx 1 - \frac{x^2}{2!}$$
$$y = 1 - x + \frac{1}{2}x^2 - \left(1 - \frac{1}{2}x^2\right) = -x + x^2 = x(x-1)$$

When $x = 0.4$, $y = 0.4 \times -0.6 = -0.24$



39)

The sum of two functions uniquely one of which must either be e^x or $\cos x$ as it goes through $(0, 1)$

So consider a point after $x \approx 2.3376$, e.g. $x = 2.4$ for the possible functions

But this is not valid for $\ln(1 + x)$ as $-1 < x \leq 1$

	4th term	$x = 2.4$	Sum of first 4 terms
e^x	$+\frac{x^3}{3!}$	2.304	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} = 8.584$
$\cos x$	$-\frac{x^6}{6!}$	−0.2654	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} = -0.7630208\ldots$
$\sin x$	$-\frac{x^7}{7!}$	−0.091	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} = 0.668550289\ldots$

$\therefore \cos x + \sin x$ is the only negative sum

So $\cos x$ and $\sin x$ must be f(x) and g(x)

40)

Area is $\frac{1}{2}ab\sin C = \frac{1}{2} \times 2 \times 3 \times \sin(2x)$

$$\sin 2x = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots \approx 2x - \frac{4}{3}x^3$$

Area is $\frac{1}{2} \times 2 \times 3 \times \left(2x - \frac{4}{3}x^3\right) = 6x - 4x^3$

$p = 6$ and $q = 4$

41 a)

$$\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots$$

$$(1+x)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)x + \frac{(-\frac{1}{2})(-\frac{3}{2})x^2}{2} + \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})x^3}{6} + \cdots$$

$$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots$$

$$\frac{\cos x}{\sqrt{1+x}} = \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots\right)\left(1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots\right)$$

$$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots - \frac{1}{2}x^2 + \frac{1}{4}x^3 - \cdots$$

$$\approx 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3$$

41 b)

$$y \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3$$

$$\frac{dy}{dx} \approx -\frac{1}{2} - \frac{1}{4}x - \frac{3}{16}x^2$$

When $x = 0$, $y = 1$

$$\left.\frac{dy}{dx}\right|_{x=0} = -\frac{1}{2}$$

$$y = -\frac{1}{2}x + 1$$

42 a)

$$\sin(2x) = 2x - \frac{(2x)^3}{3!} + \cdots = 2x - \frac{4}{3}x^3 + \cdots$$

$$\sin^2(2x) = \left(2x - \frac{4}{3}x^3 + \cdots\right)\left(2x - \frac{4}{3}x^3 + \cdots\right)$$

$$= 4x^2 - \frac{8}{3}x^4 + \cdots - \frac{8}{3}x^3 + \cdots$$

$$\approx 4x^2 - \frac{16}{3}x^4$$

42 b)

$16x^3 = 4x^2 - \frac{16}{3}x^4$ gives $4x^4 + 12x^3 - 3x^2 = 0$

$$x^2(4x^2 + 12x - 3) = 0$$

So $x = 0$, $x = -\frac{3}{2} \pm \sqrt{3}$

$$\sin^2(2x) \geq 0$$

$x = 0.232$ (3 SF)